

Evidence-Centered Visualization for ML Security Engineering

Aggregating Repository-Level Measurements Into Traceable Technical Dashboards

github.com/poojakira/mlsec-dashboards (public, default branch main, primary language Python). MIT â€œ Python 3.12 â€œ source snapshot a14ee69 â€œ verified 2026-10-01

EVIDENCE POSTURE

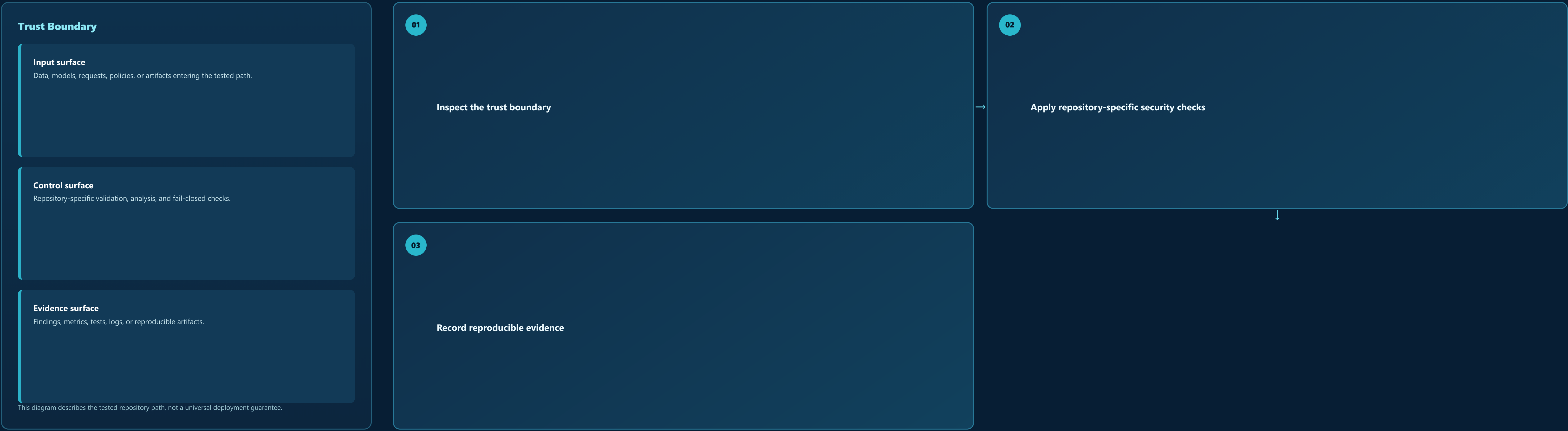
Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.



SYSTEM VIEW

Architecture and Threat-Control Flow



Research Contribution

An evidence-centered dashboard server that aggregates committed per-repo JSON into traceable HTML + a token-authed REST API, deliberately surfacing weak results (e.g. ROC-AUC 0.54, OOD F1 0.72) alongside strong ones rather than curating highlights.

Verified Evidence

EV-01	OOD F1 (redteam): 72	EV-02	Weak ROC-AUC (poison): 54
EV-03	VERIFIED_AT_SNAPSHOT â€œ Aggregates evidence from 8 active ML security repos â€œ README: dashboard_server.py discovers sibling evidence/*.json.	EV-04	VERIFIED_AT_SNAPSHOT â€œ 37 test functions â€œ Counted def test_ definitions in tests/ at source snapshot a14ee69; local verification records 37 passing pytest tests.
EV-05	VERIFIED_AT_SNAPSHOT â€œ Shows weak results explicitly (OOD F1 0.72, ROC-AUC 0.54) â€œ README states weak-alongside-strong design; numbers sourced from sibling evidence files.	EV-06	VERIFIED_AT_SNAPSHOT â€œ Token-authed REST API + static HTML â€œ README features: dashboard_server.py.
EV-07	UNSUPPORTED (disclaimed) â€œ Live SOC monitoring / real-time alerting â€œ README: static evidence snapshots, not live monitoring. Not claimed.		

Method and Control Path

- 01Inspect the trust boundary
- 02Apply repository-specific security checks
- 03Record reproducible evidence

Limitations and Residual Risk

- Static evidence snapshots, not live monitoring.
- Data can lag the current CI state of each repo.
- No alerting or real-time telemetry.
- Depends on sibling repos committing evidence.
- Local aggregation; not a hardened service.

Reproducibility

Evidence: dashboard_server.py, sibling evidence/*.json, tests/

Evidence Provenance

Claim discipline
Metrics come from the repository's verified metrics and research brief.
Scope discipline
Benchmarks are reported only at their documented fixture, dataset, or test scope.
Reproduction
PDF and PNG are generated locally from committed evidence without paid services.